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Ruei-Che Chang

Education

2022–Present **University of Michigan**, *Ann Arbor, Michigan*
Ph.D. Student in Computer Science & Engineering.
Human-AI Lab, advised by Anhong Guo.

2020–2022 **National Taiwan University**, *Taipei, Taiwan*
Visiting Student and Research Assistant.
Interactive Graphics Lab, advised by Robin Bing-Yu Chen.

2019–2021 **Dartmouth College**, *Hanover, New Hampshire*
M.S. in Computer Science.
Thesis: Adaptive Add-on Circuit Components for People who are Blind or Have Low Vision.
XDiscovery lab, advised by Xing-Dong Yang.

2014–2018 **National Cheng Kung University**, *Tainan, Taiwan*
B.S. in Electrical Engineering.

Peer-Reviewed Conference Papers

2022 [C.8] **Ruei-Che Chang**, Chao-Hsien Ting, Chia-Sheng Hung, Wan-Chen Lee, Liang-Jin Chen, Yu-Tzu Chao, Bing-Yu Chen, Anhong Guo. "OmniScribe: Authoring Immersive Audio Descriptions for 360° Videos." *In The 35th Annual ACM Symposium on User Interface Software and Technology* (**UIST'22**). Bend, Oregon. 2022. [Acceptance Rate: 26.3%]

[C.7] Ching-Wen Hung, **Ruei-Che Chang**, Hong-Sheng Chen, Chung-Han Liang, Liwei Chan, Bing-Yu Chen. "Puppeteer: Manipulating Human Avatar Actions with Intuitive Hand Gestures and Upper Body Postures." *In The 28th Annual ACM Symposium on Virtual Reality Software and Technology* (**VRST'22**). Tsukuba, Japan. 2022. [Acceptance Rate: 26.7%]

2021 [C.6] **Ruei-Che Chang**, Chih-An Tsao, Fang-Ying Liao, Seraphina Yong, Tom Yeh, Bing-Yu Chen. "Daedalus in the Dark: Designing for Non-Visual Accessible Construction of Laser-Cut Architecture." *In The 34th Annual ACM Symposium on User Interface Software and Technology* (**UIST'21**). Virtual Event. 2021. [Acceptance Rate: 21%]

[C.5] **Ruei-Che Chang***, Wen-Ping Wang*, Chi-Huan Chiang, Te-Yen Wu, Zheer Xu, Justin Luo, Bing-Yu Chen, Xing-Dong Yang. "AccessibleCircuits: Adaptive Add-On Circuit Components for People with Blindness or Low Vision." *In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (**CHI'21**). Virtual Event, Japan. 2021. [Acceptance Rate: 26.3%]

2020 [C.4] **Ruei-Che Chang***, Chi-Huan Chiang*, Shuo-wen Hsu, Chih-Yun Yang, Da-Yuan Huang, Bing-Yu Chen. 2020. "TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays." *In Symposium on Spatial User Interaction* (**SUI'20**). Virtual Event. 2020. [Acceptance Rate: 31%]

[C.3] **Ruei-Che Chang***, Yi-Shyuan Chiang*, Yi-Lin Chuang, Shih-Ya Chou, Hao-Ping Lee, I-Ju Lin, Jian Hua Jiang Chen, Yung-Ju Chang. "Exploring the Design Space of User-System Communication for Smart home Routine Assistants." *In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (**CHI'20**). Virtual Event. 2020. [Acceptance Rate: 24.3%]

[C.2] Kai-Chieh Huang, Chen-Kuo Sun, Da-Yuan Huang, Yu-Chun Chen, **Ruei-Che Chang**, Shuo-wen Hsu, Chih-Yun Yang, Bing-Yu Chen. "Glissade: Generating Balance Shifting Feedback to Facilitate Auxiliary Digital Pen Input." *In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems.* (**CHI'20**). Virtual Event. 2020. [Acceptance Rate: 24.3%] **[Best Paper Honorable Mention, Top 5% of 3126 submissions]**

2019 [C.1] Chi Wang, Da-Yuan Huang, Shuo-Wen Hsu, Chu-En Hou, Yeu-Luen Chiu, **Ruei-Che Chang**, Jo-Yu Lo, Bing-Yu Chen. "Masque: Exploring Lateral Skin Stretch Feedback on the Face with Head-Mounted Displays." *In Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (**UIST'19**). New Orleans, LA. 2019. [Acceptance Rate: 24.4%]

Posters, Demos, Works in Progress, and Extended Abstracts

[A.1] Ching-Wen Hung, **Ruei-Che Chang**, Hong-Sheng Chen, Chung-Han Liang, Liwei Chan, Bing-Yu Chen. "Puppeteer: Manipulating Human Avatar Actions with Intuitive Hand Gestures and Upper Body Postures." *In The 35th Annual ACM Symposium on User Interface Software and Technology* (**To appear at UIST Poster 2022**). Bend, Oregon. 2022.

Awards

2022 Rackham Travel Grant Award for UIST 2022.

2020 Best Paper Honorable Mention for Glissade [C.2] at ACM CHI 2020. [Top 5% of 3126 submissions]

2018 Taiwan internship scholarship from the Ministry of Education (21.53% acceptance rate).

2017 2nd Prize in Taiwan Intelligent Robot Contest (30 entries).

Research Experience

Sep 2022 – present **Human-AI Lab, University of Michigan**
Graduate Research Assistant, advised by Anhong Guo
- Developing strategy to debug customized visual program.

Sep 2021 – Apr 2022 **Human-AI Lab, University of Michigan**
Remote research intern, advised by Anhong Guo
- Developed OmniScribe for authoring immersive audio descriptions for 360° videos [C.8].

Jun 2020 – Jul 2022 **Interactive Graphics Lab, National Taiwan University**
Research Assistant, advised by Robin Bing-Yu Chen and Tom Yeh
- Developed Daedalus for non-visual accessible construction of laser-cut architecture [C.6].

Feb 2019 – Aug 2019 ***Research Assistant, advised by Robin Bing-Yu Chen and Da-Yuan Huang***
Developed Unity applications for Glissade[C.2] and Masque [C.1].
- Developed TanGo for expressive haptic interaction on VR headset [C.4].

Sep 2019 – Sep 2020 **XDiscovery Lab, Dartmouth College**
Graduate student, advised by Xing-Dong Yang
- Developed AccessibleCircuits, a set of 3D-printed add-ons adaptive to the existing circuit components and tools [C.5].

Sep 2018 – Apr 2019 **Mobile and Ubiquitous Interaction Lab, National Yang Ming Chiao Tung University**
Research Assistant, advised by Yung-Ju (Stanley) Chang
- Designed and conducted an experiment to understand the communication between human and smart home agent [C.3].

Work Experience

2018 **Mandatory Military Service**, Taiwan.

2018 **Undergraduate research intern** at *Industrial Technology Research Institute* (*ITRI*), Hsinchu, Taiwan.

Reviewer

CHI '22 '23

UIST '21 '22

CSCW '23

TEI '23

SUI '23

DIS '22

ISS '22

MobileHCI	'22
IEEE VR	'23
LBW at CHI	'20 '21 '22

Student Volunteer

UIST	'22

Student Mentoring

2022–present	**Wan-Chen Lee**, *Master student* at National Taiwan University. (Project TacNote)
2021–2022	**Fang-Ying Liao**, *Master student* at National Taiwan University. (Project BotSummoner)
2021–2022	**Chao-Hsien Ting**, *Master student* at National Taiwan University. (OmniScribe [C.7], UIST'22)
Jan–Apr 2021	**Chih-An Tsao**, *Master student* at National Taiwan University. (Daedalus [C.6], UIST'21)

Skills

Programming	Java, C#, Python, JavaScript, ROS, OpenCV, MongoDB, Swift, LaTeX
Prototyping	Arduino, 3D-printing, Fusion 360, Laser-cutting
Platforms/IDE	Unity3D, Android Studio, Fusion 360, Xcode

References

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